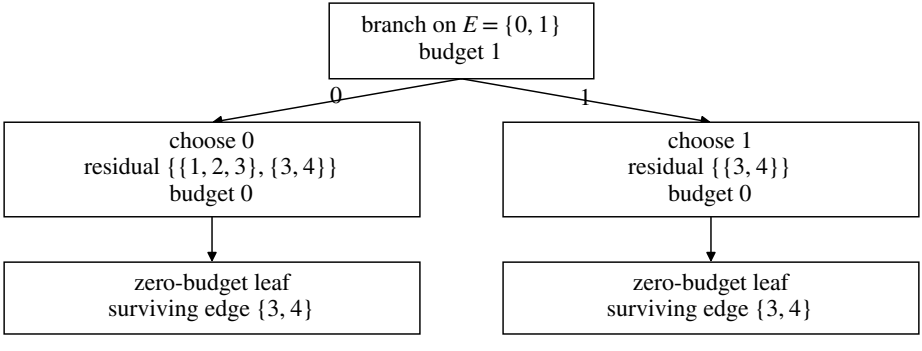
branch on $E = \{0, 1\}$
budget 1
0
1
choose 0
residual $\{\{1, 2, 3\}, \{3, 4\}\}$
budget 0
choose 1
residual $\{\{3, 4\}\}$
budget 0
zero-budget leaf
surviving edge $\{3, 4\}$
zero-budget leaf
surviving edge $\{3, 4\}$